A metaplectic Jacquet–Langlands transfer for division algebras of prime degree, I
Revision Log: Draft 8 to Draft 9 (exposition only)

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Scope

This is an expository revision requested by David Joyner, the first of four (one per Part). **No mathematical change was made**: no statement, proof, hypothesis or status label was added, removed or altered. A line-by-line `diff` of Draft 8 and Draft 9 shows changes only in section and subsection lead-ins, in the pointers of §§1.4–1.6, in the old Sections 10 (Glossary) and 11 (Literature), which were replaced, and in the bibliography, which was extended.

Numbering is unchanged. Parts II–IV and the companion paper cite Part I by number (for example "Part I, Lemma 2.2", "Proposition 2.3(b)", "Proposition 3.5"). The new material is therefore placed so that no number moves: the Summary of results is an unnumbered section between §1 and §2, the shared glossary replaces the old §10 (Glossary) under the same number, and the full literature review replaces the old §11. We compiled Drafts 8 and 9 and compared the label tables in their `.aux` files: every section, subsection, theorem-like environment, equation and appendix number is identical.

Changes

1. **(a) Shared glossary.** New file `glossary.tex` at the top of the repository, written once for the whole series and the companion paper. It uses only LaTeX, `amsmath` and `amssymb` (no Part's macros) and labels with the prefix `gl:`. Subsections: status labels (with the rule that labels are never upgraded without proof and that changes are logged; it defines **[P]**, **[E]**, **[O]**, **[C]** with the wording of Parts I–IV, and **[P^{c}]** with the wording of the companion paper); global data and the running example; local fields and the normalization $|p|_p = p^{-1}$; Hilbert symbols; covers and the cocycle normalization (Kazhdan–Patterson cocycle, parameter c before and c_v after the relabeling $\zeta \mapsto \zeta^{r'}$, the ambient cocycle, restriction as an isomorphism of extensions, scalar centrality, the global normalization by κ_w and the splittings, the star map, and the diagonal commutator matrix of Part IV, §2.1); genuine objects, central characters and good elements; orbital integrals and matching (including Part III's notation $J = \Delta O$); the Haar-measure conventions of Parts I, II and III, which differ and are listed side by side; the transform L_D with b_D and c_D; and the named results and hypotheses (Theorems A–E, C′, (H$_D$), (H), (H$'_D$), (H′), (D1)–(D4), Targets A–C). Every pointer into Parts II and III was checked against the label tables of their current

drafts (Part II, Draft 8: Definitions 2.1 and 6.4, §6.3; Part III, Draft 9: Definitions 2.1 and 4.1, Theorem 4.3).

In Part I, §10 is now "Notation and conventions": a lead-in, `\input{../glossary}`, and §10.11 "Notation specific to Part I" (about twenty entries with pointers). The entries of the old informal glossary are all kept: series-wide ones (cover, genuine, ambient cover, good element, elliptic regular element, canonical identification, normalized orbital integral, matching, supercusp form, formal degree, germ) in the shared file, and Part-specific ones (near-equivalence packet, corner Hecke algebra, elliptic detectability) in §10.11. Part I must now be compiled from `part-I/` so that `../glossary.tex` is found; the repository README was updated accordingly.

2. **(b) Summary of results.** New one-page unnumbered section after §1 (it appears in the table of contents). Seven items with labels and pointers: the ambient cover; Theorem A; Theorem B and the failure near the center; rational lifts and the trace formulas; the packet identity; Theorem C, its converse and Lemma 7.8; and (D1)–(D4) with their status in Parts II–III; plus one line on the running example. The labels are those already in the paper.

3. **(c) Literature review.** §11 was rewritten as a review for the whole series, in five paragraphs: the Jacquet–Langlands correspondence; covering groups; metaplectic correspondences; central extensions of division algebras; and the present series. Parts II–IV will cite it and add only what is specific to them. The claims of the old §11 are kept (in particular the zbMATH search over the papers citing [FK]). Twenty-six references were added: BD, FK88, Fli80, Gel76, GG18, GGW, JL, Kaz84, Klo86a, Klo86b, Kub67, Kub69, Li1, Li2, Mez01, Mez02, Mez04, PRa88, PRp96, Rap88, Rog83, Sav88, Shi73, Wal80, Wal91, Wei18. The new text cites no theorem numbers from them.

   Reference checks (bibliographic data only).

   - Checked this session against numdam/EuDML/publisher pages: Moore (Publ. IHÉS 35 (1968), 5–70; Part I's entry is *correct*), Prasad–Rapinchuk (84 (1996), 91–187), Li II (Ann. ENS 45 (2012), 787–859), Flicker–Kazhdan 1988 (J. Analyse 50, 189–200), Mezo I (CJM 54 (2002), 92–137), Mezo II (Represent. Theory 5 (2001), 524–580), Shimura (Ann. 97 (1973), 440–481), Gelbart (LNM 530, 1976), Gan–Gao and Gan–Gao–Weissman (Astérisque 398, 187–275 and 1–31), Klose 1986a (Math. Z. 191, 261–282).
   - Checked only through a citing bibliography (the reference lists of Prasad–Rapinchuk, Mezo or Moore on numdam and the publishers' pages): Prasad–Raghunathan 1988, Rapinchuk 1988, Kazhdan 1984, Flicker 1980, Kubota 1967.
   - Klose 1986b: numdam/geodesic gives pages 625–650, Prasad–Rapinchuk's list gives 625–649; we used 625–650.
   - Mezo 2004 (Pacific J. Math. 215, no. 1): the first page is 85, but we could not confirm the last page, so the entry gives no page range.
   - Li I: we could not confirm the journal; the entry cites arXiv:1004.4011.
   - Weissman (Astérisque 398): first page 33 from the volume's table of contents; last page 186 inferred from Gan–Gao starting at 187.
   - Taken from the series' existing bibliographies without a new check: Brylinski–Deligne, Jacquet–Langlands, Kubota 1969, Rogawski, Savin, Waldspurger 1980 and 1991.
   - Contents: the descriptions of the new references are at the level of their titles, abstracts and front matter. The originals of Klose's papers, Savin, and Mezo II were not available to us.

4. **(d) Lead-ins.** Every section (§§1–9 and Appendices A, B) now says, in two to four sentences, what it proves, what it depends on and what notation it introduces; §§10 and 11 have new lead-ins. Subsection lead-ins were extended where the subsection introduces notation or a hypothesis used later (§§2.2, 3.4, 3.5, 4.1, 4.3, 5.1, 5.3, 6.3, 7.1). The existing one-sentence lead-ins of the other subsections were kept as they were.

5. **Introduction.** The §1 lead-in points to the Summary and to §10; §1.4 (Conventions) and §1.5 (labels) point to the matching subsections of §10; §1.6 (Organization) describes the new §§10–11 and the Summary. The text of §§1.4–1.5 is otherwise unchanged.

6. Title line: Draft 9, November 8, 2026. Authors and acknowledgments unchanged.

Points for David (not acted on)

1. **Meaning of [O].** The request describes **[O]** as "open". Parts I–IV, the companion paper and the README all define it as "proof outlined but not fully established", and in Part I open goals are labeled **[C]**. The glossary keeps the papers' wording; changing it would relabel statements, so we have not done so.

2. **Flicker–Kazhdan, "A simple trace formula" (1988).** Its abstract says it extends the metaplectic correspondence, as well as the simple-algebras correspondence, to automorphic forms with a single supercuspidal component. This may overlap Proposition 6.2 (our variant of [FK, §18] with one supercusp form). Whether to cite it as an alternative source, or to compare the two in a remark, is a mathematical decision; the literature review cites it for its approach only.

3. **Klose's papers.** Klose (a student of Patterson) studied metaplectic extensions of quaternion division algebras (Math. Z. 191 and 193, 1986). This is prior work on covers of $D^\times$. The companion paper's literature section says that we are "not aware of an independent literature on n-fold covers of division algebras beyond [JC] and this series"; that sentence should be revised when the companion paper is next edited. Whether Klose's covers agree with those of [JC] for $r = 2$ has not been checked.

4. **Companion paper bibliography.** Its entry for Moore gives pages 157–222; the correct range is 5–70.

5. **Subsection lead-ins.** The request asks for subsection lead-ins only where notation or a hypothesis is introduced. Part I already had a one-sentence lead-in for every subsection (an earlier request); we kept them. They can be pruned if you prefer.

6. **Placement.** The Summary is unnumbered and the glossary sits at §10 (not at the front) only to keep Part I's numbering fixed for Parts II–IV. If you would rather have the glossary at the front, the cross-references in Parts II–IV would have to be updated in the same pass.

Files

`glossary.tex` (new, repository root); `part-I/mjl_part1_draft9.tex` and `.pdf`; this log (`part-I/mjl_part1_revi` and `.pdf`); `README.md` (Part I draft number, build note, labels table). Draft 8 is kept unchanged.
No scripts were added or changed.